

PREDICTCARE.AI STRATEGIC DATABASE AUDIT

UNOS / OPTN Transplant Database Audit

Data assets, modernization signals, and customer-facing growth opportunities

Prepared for UNOS strategic growth discussion

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Executive thesis

UNOS / OPTN sits on one of the richest longitudinal healthcare datasets in the United States. The opportunity is not to sell data; it is to package permissioned, de-identified, aggregated, human-governed transplant intelligence into workflow-native products that help transplant hospitals, OPOs, labs, device companies, logistics partners, payers, and policymakers increase safe organ acceptance, reduce nonuse, improve outcomes, and lower administrative burden.

Executive thesis

UNOS / OPTN sits on one of the most consequential longitudinal healthcare datasets in the United States: a national, mandatory, census-level record of solid-organ transplant candidates, donors, offers, transplants, follow-up, immunology, organ disposition, center activity, OPO activity, and outcomes.

The research-access layer is valuable, but the highest-growth opportunity is in the operational layer: real-time waitlist activity, DonorNet offer workflows, match-run behavior, acceptance and decline patterns, logistics, center benchmarking, OPO yield, data-quality issues, and predictive decision support.

The strategic opportunity is not to “sell data.” It is to package permissioned, de-identified, aggregated, human-governed transplant intelligence into workflow-native products that help transplant hospitals, OPOs, labs, device companies, payers, logistics partners, and policymakers increase safe organ acceptance, reduce nonuse, improve outcomes, and lower administrative burden.

1. Database age and strategic context

The OPTN database traces its modern operating roots to the National Organ Transplant Act and HRSA’s OPTN contract history. UNOS was awarded the OPTN contract in 1986, and SRTR Standard Analysis Files broadly include OPTN-collected elements on U.S. solid-organ transplant candidates, donors, and recipients from **1987 to present**.

Year	Milestone	Strategic implication
1984	National Organ Transplant Act	Federal framework for national organ allocation and oversight
1986	HRSA awarded OPTN contract to UNOS	National operating system for transplant matching and data collection
1987-present	SRTR/OPTN research files	Nearly four decades of transplant candidate, donor, recipient, and outcome history
2016	UNOS API/developer program expanded through Apigee	Shift from manual data entry toward real-time data exchange
2024-2025	HRSA OPTN modernization / multi-vendor transition	Opening for next-generation IT, interoperability, transparency, analytics, and customer-facing growth products
2025+	Next-generation OPTN IT modernization	Opportunity to build a flexible, longitudinal, AI-ready transplant data ecosystem

2. Core UNOS / OPTN data systems

UNOS historically operated the OPTN technology platform, **UNet**, with several major modules.

System / module	What it captures	Commercial opportunity
Waitlist	Candidate registration, active/inactive status, organ-specific urgency, listing details, clinical status, demographics	Candidate journey analytics, referral conversion, waitlist risk, access/equity analytics
DonorNet	Deceased donor data, OPO donor management, match runs, organ offers, attachments/images, offer review, acceptance/decline workflow	Offer acceptance optimization, nonuse root-cause analytics, center decision support
TIEDI	Pre- and post-transplant forms: candidate, recipient, donor, living donor, follow-up, immunosuppression	Data-quality automation, audit readiness, outcome intelligence
TransNet	Organ packaging, labeling, transport validation, recipient-organ identity confirmation	Logistics optimization, ischemic-time reduction, transport performance analytics
UNet APIs	Data exchange with OPOs, transplant programs, labs, EHR/EMR systems, donor systems	Workflow integration, partner ecosystem, marketplace/API products
UNOS Tools / Data Services	Dashboards, benchmarking, reports, quality and process improvement tools	Premium analytics, customer success, strategic advisory, subscription products

3. What is in the research-access / public data layer

The SRTR public Standard Analysis Files are organized by candidate, donor, transplant, follow-up, histocompatibility, immunosuppression, institution, and organ-specific tables. They are released quarterly and currently distributed in SAS format.

Major file groups

File group	Example tables	What they contain
Candidate / waitlist	CAND_KIPA, CAND_LIIN, CAND_THOR	Candidate demographics, listing date, organ type, blood type, diagnosis, clinical status, comorbidities, height/weight/BMI, dialysis, urgency fields, donor acceptability criteria
Status history	STATHIST_KIPA, STATHIST_LIIN, STATHIST_THOR	Active/inactive periods, priority status, status reasons, MELD/LAS/CAS-related variables, status justification history
Deceased donor	DONOR_DECEASED	Donor ID, ABO, age, cause of death, DCD/DBD-related data, labs, serologies, HLA, risk factors, infectious disease, organ-specific donor details
Living donor	DONOR_LIVE, DON_LIV_FOL	Living donor demographics, ABO/HLA, pre/post-op labs, complications, donor follow-up at 6 months, 1 year, and 2 years
Donor organ disposition	DONOR_DISPOSITION	Organ recovered/transplanted/discarded, discard reason, recovery date, share type, storage/preservation solution, match ID
Transplant event	TX_KI, TX_LI, TX_HR, TX_LU, TX_PA, TX_KP	Recipient/candidate + donor + transplant event summary, center/OPO, dates, organ type, ischemic time, diagnosis, prior transplant, risk factors
Transplant follow-up	TXF_KI, TXF_LI, TXF_HR, TXF_LU, etc.	Survival, graft failure, rejection, infection, malignancy, dialysis after transplant, functional status, cause of death
Histocompatibility	REC_HISTO, REC_HISTO_XMAT, PRA_HIST	Donor/recipient HLA, PRA/CPRA, antibody screening, crossmatch method/result, typing method
Immunosuppression	IMMUNO, FOL_IMMUNO	Induction, maintenance, anti-rejection drugs, drug codes
Malignancy / treatment	MALIG, TREATMENT	Post-transplant malignancy diagnosis and treatment
Institutions	INSTITUTION, HIST_OPO_TXC	Transplant center / OPO identifiers, service relationships, historical mappings

Example variable scale from the SRTR public dictionary

Data file	Approx. variable count
DONOR_DECEASED	260
DONOR_LIVE	233
DONOR_DISPOSITION	17
CAND_KIPA — kidney/pancreas candidates	194
CAND_LIIN — liver/intestine candidates	174
CAND_THOR — heart/lung candidates	213
TX_KI — kidney transplant	283
TX_LI — liver transplant	307
TX_HR — heart transplant	327
TX_LU — lung transplant	313
TXF_KI — kidney follow-up	106
REC_HISTO	111
REC_HISTO_XMAT	8
PRA_HIST	10

4. Practical data-field inventory

Candidate / recipient data

The database includes, or has historically included, fields such as:

- Identity and operational fields in restricted systems: name, DOB, SSN, center patient ID, transplant center, organ listed
- Demographics: sex/birth sex/gender depending on form era, age, race, ethnicity, citizenship, state and ZIP of permanent residence
- Listing and waitlist: listing date, activation date, active/inactive status, wait time, reason for removal, death date when applicable
- Clinical baseline: ABO, height, weight, BMI, primary diagnosis, secondary diagnosis, functional status, education/employment fields in some forms
- Comorbidities: diabetes, dialysis, peripheral vascular disease, cerebrovascular disease, COPD, hypertension, malignancy, previous transplant history
- Organ-specific clinical variables: MELD/PELD, Lung CAS/LAS-era fields, heart status, kidney allocation variables, CPRA/PRA, liver exceptions, thoracic status justifications
- Donor acceptability: HIV-positive organ willingness, HBV core antibody positive, HCV positive, DCD, A2/A2B, ABO-incompatible or other organ-specific criteria

Donor data

The donor layer includes:

- Donor ID, donor type, OPO, donor hospital context
- ABO, age, sex, race/ethnicity, citizenship
- Cause of death, DCD status, stroke/CVA, cardiac arrest, donor-risk history
- Serologies and infectious disease fields: CMV, EBV, HIV, HBV, HCV, HTLV, Chagas NAT in some eras/contexts
- HLA: A, B, C, DR, DQ, DP, Bw4/Bw6
- Labs and clinical markers: creatinine, BUN, bilirubin, liver enzymes, albumin, organ-specific markers
- Risk factors: hypertension, diabetes, cancer history, tobacco, alcohol, cocaine, IV drug use
- Donor management and recovery: clamp date/time, recovery date/time, consent/authorization fields, organ-specific evaluation, biopsy, anatomy, storage solution

Matching / compatibility data

The matching layer includes:

- ABO compatibility
- Donor and recipient HLA typing
- CPRA/PRA and unacceptable antigen data
- Crossmatch source, type, technique, result, auto-crossmatch result
- Candidate acceptability filters and organ-specific thresholds
- Medical urgency and priority status
- Wait time
- Pediatric priority
- prior living donor / donation status priority
- geographic/distance allocation inputs
- organ-specific allocation scores

Transplant event data

The transplant-event layer includes:

- Transplant date and organ type
- Multi-organ transplant flags
- recipient center and OPO
- donor ID and donor type
- ischemic time
- recipient condition at transplant
- diagnosis and previous transplant history
- surgeon/NPI in restricted form data
- induction immunosuppression
- discharge and early post-transplant status

Follow-up / outcomes data

The outcomes layer includes:

- patient alive/dead status
- date last seen and date of death
- cause of death
- graft status, graft failure date, graft failure cause
- rejection episodes, biopsy-confirmed rejection
- dialysis after transplant
- disease recurrence
- malignancy and treatment
- infections and antivirals
- immunosuppression regimen
- functional status and selected social/education/employment fields
- living donor follow-up at 6 months, 1 year, and 2 years

5. What is limited, restricted, or not consistently public

The public/research-access SAF layer is powerful, but not the full commercial asset. SRTR notes that SAFs generally do not include geographic data or text fields, and patient-level ZIP codes or supplementary files may require special request, approval, and cost.

Data type	Likely availability	Why it matters
Patient name, SSN, center patient ID	Restricted operational PHI/PII	Not appropriate for commercial use except under strict permissions
Patient-level ZIP/full geography	Usually excluded or restricted	Market access, equity, logistics, referral geography
Free-text exception narratives	Restricted	NLP opportunity, but high governance burden
Real-time match-run logs	Operational / not broadly public	Highest-value offer behavior analytics
Offer acceptance/decline behavior	Partially captured; richer in DonorNet	Center decision support and nonuse reduction

Data type	Likely availability	Why it matters
Donor imaging / attachments	Operational only	AI-assisted organ quality assessment
Organ transport telemetry	Fragmented	Logistics and ischemia-risk modeling
Perfusion-device data	Often held by device companies	DCD and machine-perfusion outcomes
EHR longitudinal labs/vitals	Not consistently in OPTN today	Dynamic risk prediction
ICU / intra-op detail	Not consistently captured	Post-transplant outcome prediction
API/workflow logs	UNOS/internal	Operational bottlenecks and customer experience

6. New data likely coming into the ecosystem

Confirmed direction

UNOS and the broader OPTN modernization process are moving toward more API-based, interoperable, dynamic data collection.

UNOS API capabilities already support:

- Death notification registration
- Deceased donor registration and DCD forms
- Donor record upload and donor attachments
- Donor hospital lists
- Waitlist clinical data exchange from EHR/EMR
- Allocation calculators: CPRA, Lung CAS, MELD, PELD
- TIEDI forms: TRF, IMF, TRR, IMR, TCR, LDR, LDF

Public Google Cloud materials describe UNOS's API footprint as having hundreds of external endpoints, 300+ connected healthcare organizations, and high-volume real-time transactions across UNet.

Direction of travel

Emerging data category	Strategic value
EHR-extracted labs, vitals, hemodynamics via FHIR	Dynamic risk prediction, lower manual burden
Longitudinal candidate deterioration data	Better urgency and offer-acceptance decisions
Granular mechanical circulatory support categories	More accurate heart allocation and risk adjustment
DCD procurement workflow timing	DCD outcomes, organ viability, process improvement
Normothermic regional perfusion / direct procurement and perfusion variables	Compare DCD strategies and outcomes
Organ perfusion-device data	Device benchmarking and viability prediction
Organ transport system and temperature data	Logistics optimization and ischemia-risk modeling
Donor organ imaging	AI-assisted organ quality assessment
Histopathology of explanted organs	Diagnosis validation and outcomes research
Genetic testing for disease etiology	Risk stratification and disease-specific insights
More specific nonuse / decline reasons	Donor-pool expansion and center behavior analytics
Offer behavior / next-offer prediction	Point-of-offer decision support

Emerging data category	Strategic value
API workflow metadata	Operational analytics and customer experience improvement

7. Informed view of the non-public operational data asset

The highest-value data likely exists in the operational UNet layer. Based on UNet workflows, DonorNet, TIEDI, TransNet, and API architecture, UNOS likely has or can derive:

Match-run and offer-level intelligence

- Every match run generated
- Candidate rank/order for each organ
- Which centers were offered the organ
- Offer timestamps and response timestamps
- Accepted / declined / bypassed / no-response outcomes
- Decline reason codes and possibly richer operational context
- Whether offer details, imaging, or attachments were reviewed
- Subsequent candidate outcome after accepting or declining
- Whether the candidate received another offer later
- Time to next offer
- Whether the organ was transplanted elsewhere or discarded
- Center-specific acceptance tendencies

Center behavior intelligence

- Organ acceptance rates by center, organ type, donor risk tier, geography, and candidate profile
- Response latency
- Decline-code distribution
- Outlier behaviors versus peer programs
- Waitlist size, transplant conversion, volume trends
- Data quality errors and TIEDI completion performance
- Staffing and workflow bottlenecks
- MPSC/SRTR-related risk signals

OPO performance intelligence

- Referral-to-authorization conversion
- Donor management timelines
- Organ yield
- Recovery process timing
- DCD progression and agonal-phase timing
- Organ nonuse and discard reasons
- Donor hospital patterns

- OPO-to-center placement dynamics

Logistics intelligence

- Packaging, labeling, handoff, transport, and arrival timestamps
- Cold ischemic time and delay drivers
- transport mode and route performance
- courier/vendor performance
- validation events and exceptions

8. Current product evidence and white space

UNOS is already moving from database stewardship toward productized intelligence.

Known product/product-like examples include:

- CARE Tool — broader organ acceptance opportunities
- UNOS Predict — forward-looking metrics relative to future SRTR cohorts
- Benchmark Report — customizable patient population, listing, and transplant activity insights
- UNOS Staffing Survey — staffing benchmark for transplant administration
- Organ Utilization Tool / OPO tools
- DonorNet predictive analytics — time-to-next-offer and three-year mortality likelihood for adult kidney offers
- UNet APIs — workflow integration with OPOs, transplant programs, labs, and EHR systems
- UNOS One / Marketplace — emerging platform/partner ecosystem direction

The white space is to package these capabilities into a more explicit customer-value architecture: business outcomes, workflow outcomes, risk reduction, center growth, OPO performance, logistics performance, and responsible AI/data modernization.

9. Commercial packaging opportunities

Product concept	Buyer	Value proposition
Offer Acceptance Optimizer	Transplant centers	Identify safe acceptance opportunities where the center is more conservative than peers
Next-Best-Offer Advisor	Transplant physicians	Support accept-now vs wait decisions using candidate mortality and likely future offer patterns
OPO Yield Intelligence	OPOs	Identify where donors/organs are lost across referral, authorization, recovery, placement
Organ Nonuse Root-Cause Dashboard	OPOs, centers, HRSA	Break discard/nonuse into donor quality, center behavior, logistics, data quality, or process issues
SRTR/MPSC Early Warning	Transplant centers	Predict performance risk before public reports or regulatory scrutiny
Data Quality Copilot	Centers and OPOs	Detect missing, late, or contradictory OPTN/TIEDI data before audit
Transplant Market Access Atlas	Device, diagnostic, pharma, payer partners	De-identified market map of volume, patient mix, disease burden, organ demand, center archetypes
Perfusion Device Outcomes Registry	Device companies, centers	Link machine perfusion/device data to outcomes and adoption patterns
DCD Expansion Intelligence	OPOs, centers, device companies	Optimize DCD procurement, NRP/DPP strategy, and machine perfusion usage

Product concept	Buyer	Value proposition
Histocompatibility Intelligence	HLA labs, kidney programs	Improve sensitized-candidate strategy, CPRA trends, crossmatch outcomes
Logistics Optimization Platform	OPOs, logistics partners	Reduce cold ischemia and predict transport delay risk
Equity / Access Analytics	HRSA, centers, foundations	Identify disparities in listing, offers, wait time, acceptance, transplant rate, outcomes

10. Recommended growth wedge

Transplant Performance Command Center

Positioning: Help transplant programs increase safe organ acceptance, improve SRTR/MPSC readiness, reduce administrative burden, and find hidden growth opportunities using OPTN/SRTR-derived intelligence and workflow automation.

Core modules

1. Acceptance Opportunity Map

Identifies donor profiles a center may be declining too conservatively relative to comparable peers.

2. Waitlist Conversion Analytics

Tracks listed patients → offers → accepts → transplants → outcomes; highlights drop-off by organ, diagnosis, demographics, geography, and referral channel.

3. SRTR/MPSC Early Warning

Shows expected risk trajectory before public reports, allowing leaders to act earlier.

4. Data Quality + TIEDI Automation

Detects missing data, contradictions, late forms, and audit risks.

5. Referral / Growth Map

Finds under-penetrated referral geographies, hospitals, nephrology/cardiology/pulmonology networks, and access gaps.

6. Executive Benchmark Report

Quarterly board-ready narrative and dashboard for transplant leadership.

Potential commercial model

- Entry analytics subscription: \$5k-\$10k/month
- Enterprise command center: \$10k-\$30k/month
- Strategic advisory + custom analytics: \$25k-\$75k/project or retainer
- Partner marketplace / API monetization: usage-based or revenue-share
- Device / diagnostics registry analytics: custom enterprise agreements

11. Messaging implications for UNOS

Current implicit message

UNOS is seen primarily as the national transplant infrastructure and data steward.

Growth message opportunity

UNOS can become the trusted intelligence layer that helps every transplant stakeholder make better, faster, more equitable, more evidence-based decisions.

Suggested narrative

UNOS is uniquely positioned to transform transplant data into trusted, workflow-native intelligence that helps centers accept more organs safely, helps OPOs increase yield, helps patients receive transplants sooner, and helps the entire ecosystem learn faster from every donor, offer, transplant, and outcome.

Messaging pillars

Pillar	Plain-English meaning
Trusted national data infrastructure	UNOS has the census-level operating data others cannot replicate
Human-governed decision support	Data supports clinical judgment; it does not replace it
More transplants, safer decisions	Growth is mission-aligned: reduce nonuse, improve acceptance, protect outcomes
Workflow-native products	Insights appear where the decision is being made, not in a disconnected report
Customer maturity	Move from data collection to proactive intelligence, benchmarking, and action
Modernization with governance	AI/interoperability must be privacy-preserving, validated, auditable, and trusted

12. Compliance, trust, and governance guardrails

The market opportunity must be framed carefully.

Safe framing

- “Permissioned, de-identified, aggregated transplant intelligence”
- “Human-governed decision support”
- “Operational benchmarking and process improvement”
- “Workflow integration that reduces burden and improves data quality”
- “Mission-aligned growth that increases safe organ utilization”

Avoid

- “Selling patient data”
- “Monetizing PHI”
- “AI replacing physicians”
- “Guaranteed improved survival” without prospective validation
- Re-identification, scraping, or use of restricted data outside data-use agreements
- Overstating what is public versus restricted

Required controls for future products

- HIPAA and federal data-use compliance
- Strong de-identification / aggregation thresholds
- Data-use agreements and role-based access
- IRB/legal review where research or patient-level analysis is involved
- Model validation and bias monitoring

- Clinical disclaimers and human-in-the-loop workflows
- Audit trails and explainability for decision support

13. Recommended next steps

1. Create a product architecture map

Map current UNOS Tools, UNet APIs, DonorNet predictive analytics, OPO tools, and UNOS Marketplace into a single customer-facing product portfolio.

2. Define three flagship buyer journeys

- Transplant center administrator / quality leader - OPO executive / operations leader - Technology / device / logistics partner

3. Build a “value from data” message house

Reframe data from compliance burden to mission-aligned intelligence: more transplants, lower burden, safer decisions, better customer value.

4. Package a Transplant Performance Command Center pilot

Pick one organ area, one customer segment, and one measurable outcome: safe acceptance, nonuse reduction, or SRTR/MPSC readiness.

5. Create a governance-first AI roadmap

Position AI around validated workflow support, not hype: offer triage, missing data detection, nonuse root cause, logistics prediction, and center benchmark explanation.

6. Develop customer-facing collateral

Build a concise deck, one-page product map, case-study template, and executive demo narrative for UNOS leadership and prospective customers.

14. Cover-letter language

Recommended bullet for Croom’s cover letter:

I have also begun a structured audit of the UNOS / OPTN data ecosystem — including candidate, donor, match, transplant, follow-up, OPO, center, API, and predictive analytics assets — to identify responsible, mission-aligned growth opportunities in benchmarking, decision support, data quality, organ utilization, and customer-facing intelligence products.

Shorter version:

I have already started developing a data-informed growth thesis for UNOS: how the OPTN/UNet ecosystem can be responsibly packaged into trusted benchmarking, decision-support, organ-utilization, and customer intelligence products that advance the mission while creating durable new value for transplant stakeholders.

Appendix: Source map

Primary source categories reviewed:

- SRTR Standard Analysis Files public data dictionary
- SRTR Standard Analysis Files overview
- UNOS UNet / Waitlist / DonorNet / TIEDI / TransNet public pages
- UNOS API public pages and developer-portal signals
- UNOS predictive analytics / DonorNet kidney offer analytics materials

- Google Cloud UNOS / Apigee case study
- OPTN/TIEDI data collection form instructions from RegInfo.gov
- 2025 American Journal of Transplantation perspective: “Modernizing the national transplant database”

Important diligence caveat: some OPTN/HRSA pages were blocked by automated access controls during research. The audit therefore distinguishes between confirmed public/research-access data and informed operational inferences based on UNet/TIEDI/DonorNet workflows, public documentation, SRTR dictionaries, and modernization literature.